Product Introduction

The ZDH1562 is a high-gain, high-efficiency power amplifier. It operates from DC to 1000MHz, delivering up to 36dBm output power and 66% efficiency from a 3.7V supply. It features 50Ω input/output matching.

The ZDH1562 is packaged in a standard green lead-free QFN5x5-28PIN package, offering advantages such as low cost and high reliability.

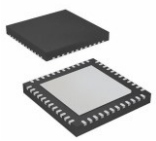
Typical application scenarios

- Mobile Radio Amplifier
- Walkie-Talkie Amplifier
- Wireless Data Transmission (FSK, AFSK, OOK, GFSK etc.)

Absolute Maximum Ratings

Parameter	Value
Storage temperature	-65°C~+150°C
Operating temperature	-55°C~+125°C
Supply voltage (V_{CC})	+6V
Supply current (I_{CC})	+5.0A
RF input power (RFIN)	+15.0dBm

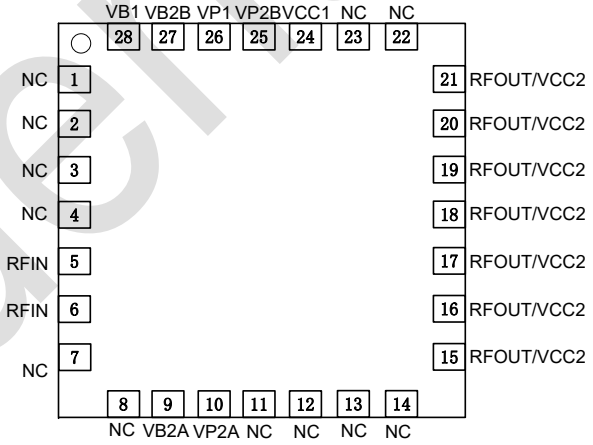
Features



- 3.7V supply voltage, 1.5A operating current
- Typical gain: 37dB @ 440MHz
- Typical P1dB: 36dBm @ 440MHz
- Typical OIP3: 40dBm @ 440MHz
- Efficiency: >65% @ 440MHz
- Input and output 50Ω matching
- Green lead-free 28-pin QFN5x5 package

This product complies with all relevant regulations and is halogen-free

Pin diagram (Top View)



PIN No.	Pin name	Description
1-4,7,8, 11-14,22,23	NC	Empty, floating or grounded
5,6	RFIN	RF input
9	VB2A	Second stage bias voltage A
10	VP2A	Second stage drive voltage A
15-21	RFOUT/VCC2	Radio frequency output/second stage power supply voltage
24	VCC1	First stage power supply voltage
25	VP2B	Second stage drive voltage B
26	VP1	First stage drive voltage
27	VB2B	Second stage bias voltage B
28	VB1	First stage bias voltage
Other	GND	GND

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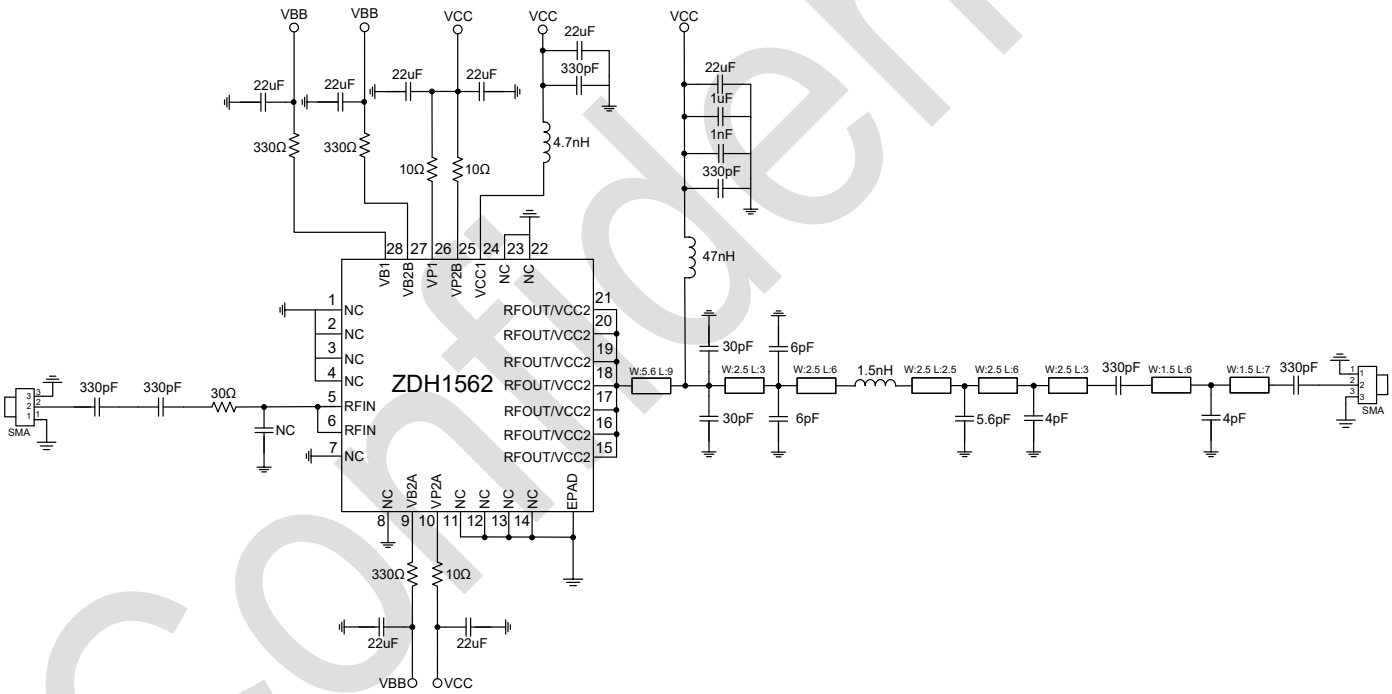
Typical application electrical parameters

Test conditions: VCC=+3.7 V, VBB=+3V, Temp=+25°C, 400MHz~510MHz application circuit, 50Ω test system.

Parameter	Value				Unit
Frequency	400	440	470	510	MHz
Small-signal gain	38	37	37	36	dB
Input return loss (S11)	-15	-15	-15	-15	dB
Output return loss (S22)	-10	-12	-13	-23	dB
P1dB	36	36	36	36	dBm
OIP3（Pout = 20dBm / tone）	40	40	39	38	dBm
Quiescent current (Icq)	712	712	712	712	mA
Operating current (Icc)	1.75	1.57	1.58	1.56	A
Efficiency (η)	54	65	66	65	%

Typical application circuit (400MHz~510MHz)



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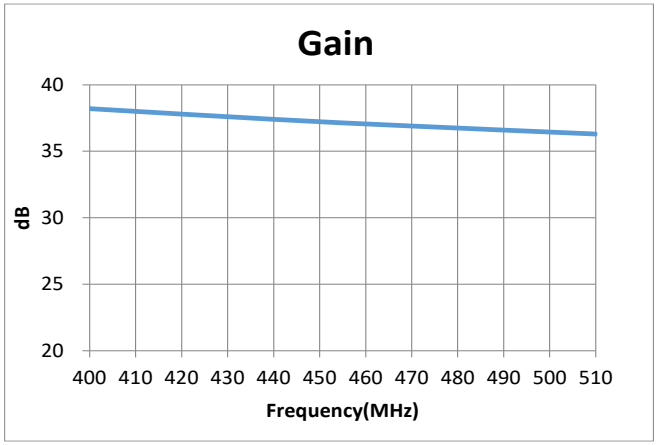
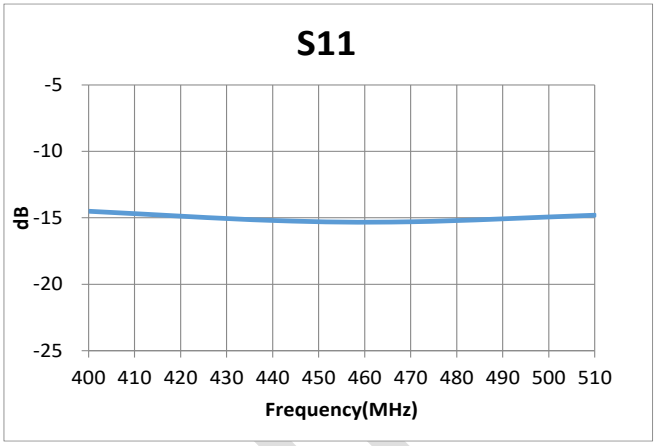
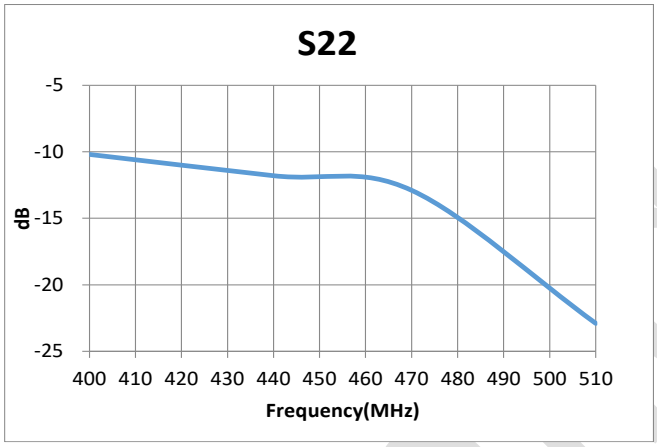
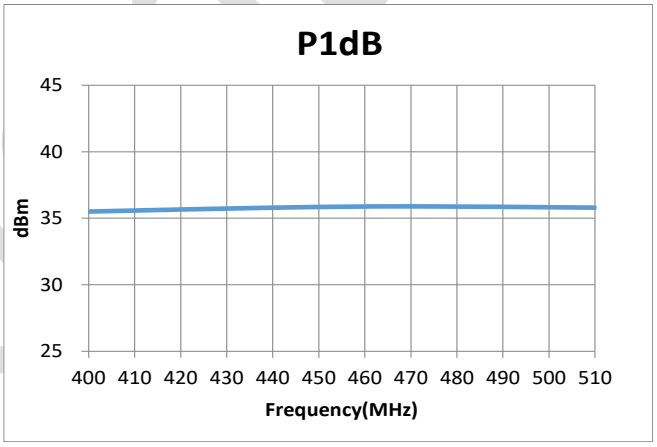
Typical performance curves

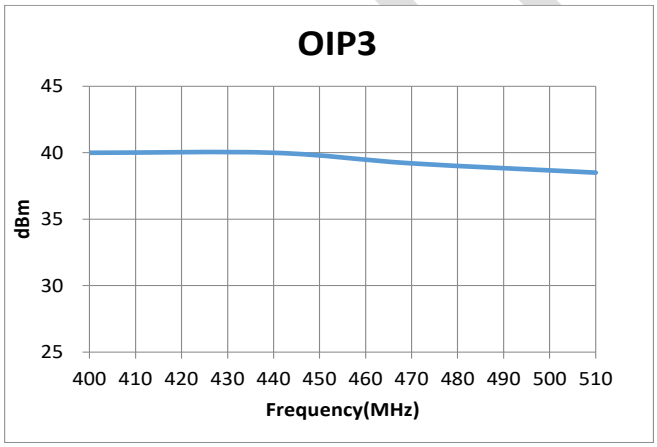
Test conditions: VCC=+3.7 V, VBB=+3V, Temp=+25°C, 400MHz~510MHz application circuit, 50Ω test system.

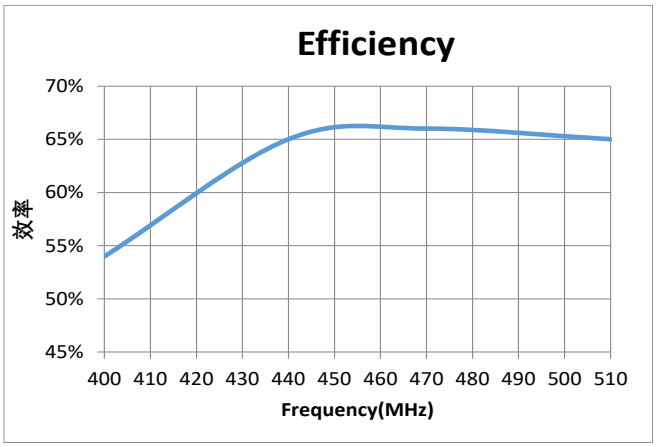




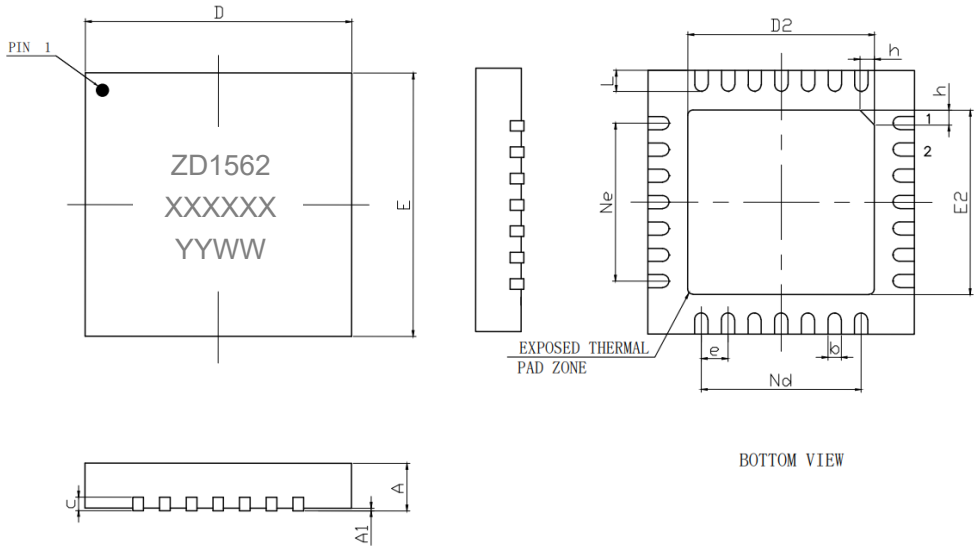








Package diagram



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0. 70	0.75	0. 80
A1	—	0.02	0. 05
b	0. 18	0.25	0. 30
c	0. 18	0.20	0. 25
D	4. 90	5. 00	5. 10
D2	3. 40	3. 50	3. 60
e	0. 50BSC		
Ne	3. 00BSC		
Nd	3. 00BSC		
E	4. 90	5. 00	5. 10
E2	3. 40	3. 50	3. 60
L	0. 35	0.40	0. 45
h	0. 30	0. 35	0. 40
L/F载体尺寸 (MIL)	150*150		

Ordering Information

Model	Marking	Packaging
ZDH1562	ZD1562	QFN5x5-28

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